GridIQ

Power-quality assessment

Demo Corporation · Sandton Mall

PROVISIONAL: the assessment is still running. Results cover the elapsed period only and are not final.

Assessment **GQ-A-0005 · 21-day assessment**	Report **revision 0 · provisional preview**
Observation period (SAST) **2026-11-01 21:00 -> 2026-11-11 21:11**	Observed **10.01 days of 21**
Site address **99 Retail St, Sandton**	Measurement points **HVAC Feed (incomer)**

1. Scope and method

Ten-minute values of every measured phase across the observation period, screened against NRS 048-2 (screening thresholds; formal compliance needs a Class A instrument, one full week and >= 95 % valid 10-minute values). Observed values, derived indicators, rule findings, possible causes and recommended verification are kept separate below.

Point	Coverage	Samples	Reporting interval	Gaps >= 30 min	Meter
HVAC Feed	100.0 %	15 713	300 s	2	device

Coverage = samples received × the meter's median reporting interval ÷ the observation window. Missing data is never filled or shown as zero.

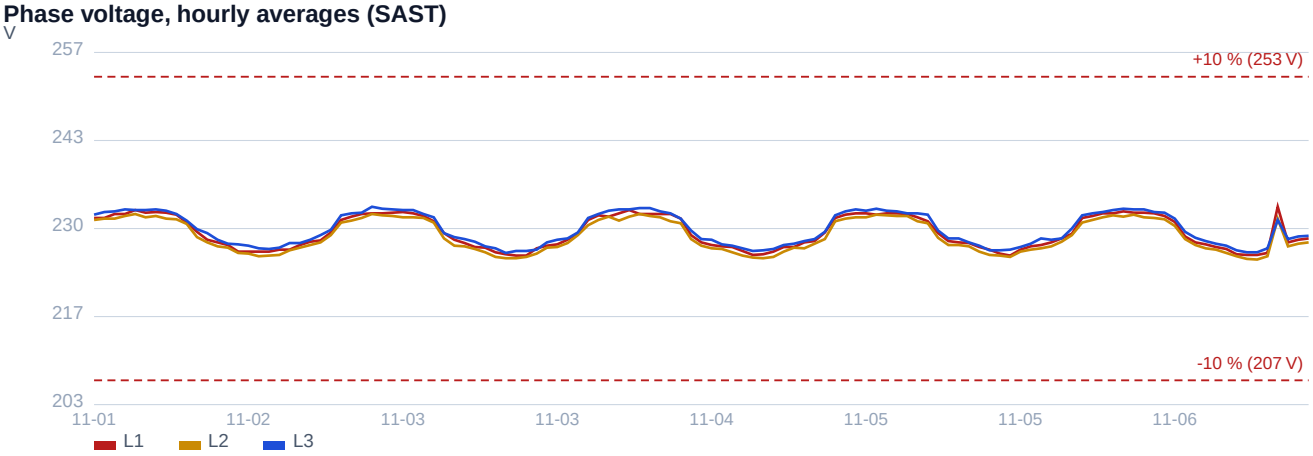
HVAC Feed

Verdict type: screening only. The meter is not confirmed as an IEC 61000-4-30 Class A instrument; values are screened against NRS 048-2 thresholds, not certified.

Voltage (observed)

Phase	Min (V)	Max (V)	P5 (V)	P95 (V)	Outside ±10 % of 230 V	10-min values
A	224,1	234,2	226	232,7	0.0 %	710
B	223,9	234,1	225,5	232,2	0.0 %	710
C	225,2	234,5	226,5	233,1	0.0 %	710

Phase voltage, hourly averages (SAST)

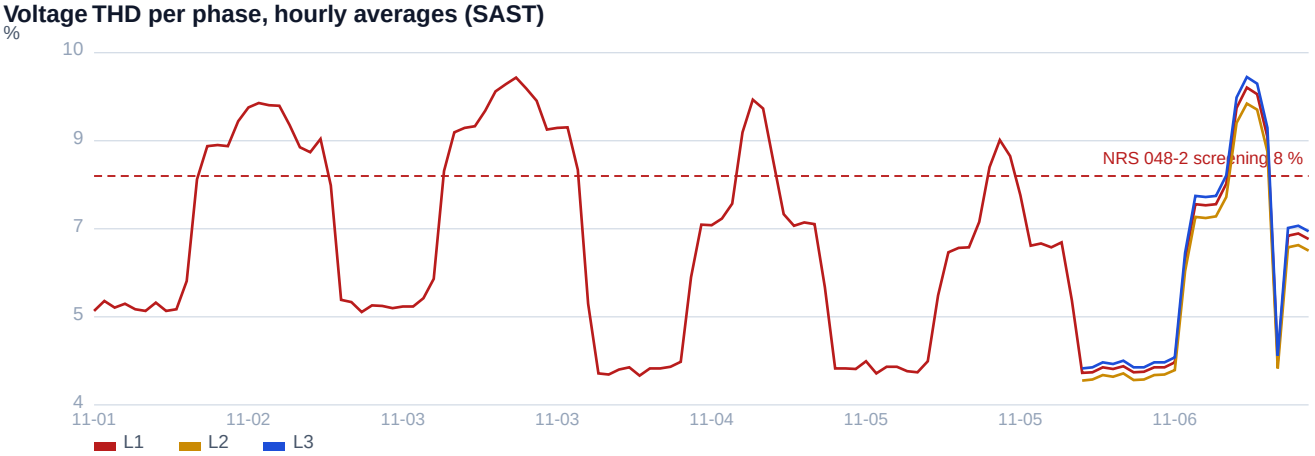


Harmonic distortion: a finding

Phase	THD-V P95 (%)	THD-V max (%)	10-min values > 8 %	Worst hours (SAST)
A	9.50	10.17	31.0 %	13:00, 14:00, 15:00
B	9.26	9.44	18.4 %	13:00, 14:00, 15:00

C	9.73	9.94	21.6 %	13:00, 14:00, 15:00

Voltage THD per phase, hourly averages (SAST)



Current THD (P95): A 41.6 %, B 43.5 %, C 41.4 %. THD-I is relative to the present current, not TDD: no IEEE 519 conclusion is drawn without the demand current and fault level.

Individual orders: Individual harmonic orders were not recorded (THD only). No spectrum is shown or inferred. To identify harmonic sources, log individual orders (e.g. 3, 5, 7, 11, 13) per phase.

Possible causes (hypotheses, to verify)

- Non-linear loads on site: variable-speed drives, rectifiers, UPS, LED lighting. Distortion that follows the site's operating hours points this way.
- Background distortion from the supply network: distortion present outside operating hours, or at a similar level on all phases, points this way.
- Resonance with PF-correction capacitors (if installed and not detuned) can amplify a harmonic order.

Recommended follow-up measurements

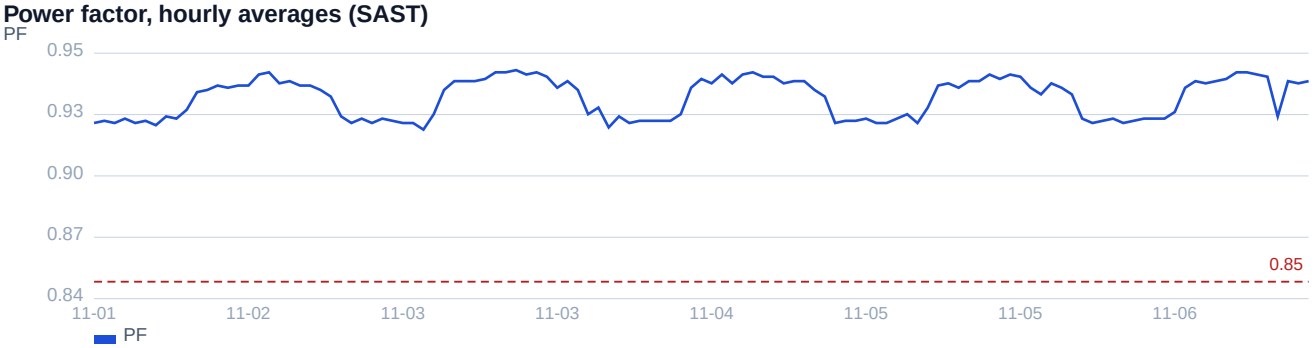
- Log individual voltage and current harmonic orders (at least 3, 5, 7, 11, 13) per phase for one week at the incomer and the main suspected feeders.
- Record the transformer rating and the fault level at the point of supply; list drives and capacitor banks with their ratings.

No filter or capacitor size is recommended from THD data alone; that needs an engineering study.

Power factor and reactive demand

Average power factor while loaded 0.933; 5th percentile 0.919; 0 % of loaded 10-minute values below 0.85. Low PF increases kVA and reactive charges; it is not a supply-quality breach.

Power factor, hourly averages (SAST)



Unbalance, frequency, interruptions and events

Voltage unbalance P95 0.44 %; current unbalance P95 1.8 % (unbalance as computed by the meter (per its configuration); not stated to be negative-sequence unbalance). Frequency 49.86 Hz–50.14 Hz; 0 10-minute value(s) outside 49–51 Hz. 0 minute(s) with voltage below 23 V.

0 excursion event(s). Source: inferred from the sampled RMS trend (pq_events), not device-recorded. device event/ waveform capture not confirmed: transient and sub-second claims are not made.

Rule findings for this point

Finding	Observed	Recommended verification	Review
Voltage distortion above the NRS 048-2 limit (THD 9.7 %)	The 95th percentile of 10-minute voltage THD was 9.7 % on phase C, above the 8 % limit (last 7 days, 25.3 h metered). Distortion overheats motors and transformers and can trip sensitive equipment.	Identify the main harmonic sources (VSDs, UPSs, LED drivers, rectifiers) with a harmonic survey, then consider active or passive filtering. Any capacitor bank on this supply should be detuned.	not yet reviewed

Technical appendix

Dataset schema
gridiq.assessment-dataset/1

Dataset SHA-256
549679499c9ac90a0a65451b344b88f1…

Calculation version
gridiq-calc-2026.11.1

Rule version
gridiq-rules-2026.11.1

Template version
gridiq-templates-2026.11.1

Data cutoff (SAST)
2026-11-11 21:11

Generated (SAST)
2026-11-11 21:11

Time convention
[start, end): a value at bucket B covers B <= t < B + interval

Method

- Readings are attributed to a measurement point by deployment window at the measurement time; buckets that straddle an install or removal are recomputed from the samples inside the window.
- Energy from cumulative kWh register differences per deployment; a negative step is a register reset and never summed through; power integration only where registers are absent.
- Billed demand: 30-minute minute-weighted kVA averages; instantaneous peaks are reported separately.
- Power quality: 10-minute values screened against NRS 048-2 (screening thresholds; formal compliance needs a Class A instrument, one full week and >= 95 % valid 10-minute values).
- Sign convention: import positive; the deployment sign correction is applied. Times in the data are UTC; this report shows SAST (UTC+2).

Full SHA-256 of the dataset: 549679499c9ac90a0a65451b344b88f14b30c0bd16e876f6f029b7a1bd2f5508